

Problem - Solution Fit Template v1

Phase 3: Project Design Phase

This phase establishes the structural blueprint of the **shopEZ** platform, defining the architectural pattern, data flows, database schemas, and problem-solution maps.

1. Problem - Solution Fit Matrix

To address the target audience's core complaints, shopEZ introduces modern MERN stack patterns:

Customer Pain Points	Existing Alternatives (e.g. Shopify, Monoliths)	shopEZ Solution Fit
Cart desync & item loss	Session-only cookies or basic local storage which clears on session timeout.	Persistent MERN Lifecycle: State is bound to React Context, syncing with local storage and database endpoints on login.
Poor catalog search/filtering	Simple string matches that ignore variants and return heavy, unfiltered datasets.	Dynamic Backend Filters: Mongoose queries support autocomplete suggestions, regex search, pricing boundaries, and rating filters.

Unoptimized media loading	Local server hosting which bottlenecks bandwidth and loads high-resolution files.	Cloudinary CDN Pipeline: Uploaded images are offloaded to Cloudinary, serving optimized images based on user viewport.
Complex administrative controls	Split admin pages or generic CMS dashboards with no real-time inventory alerts.	Unified Admin Ledger: Live analytics displays monthly earnings graphs, user toggles, and low-stock indicators.